The mathematics and the debt

Companion to strips 01–08 · supplied structure, derived consequences, explicit limits

What these pictures can establish

This is a supplied classical model with one degree of freedom. It makes a local Hamiltonian construction and one Noether example visible. It does not derive mechanics from marks alone, nor yet explain quantum mechanics.

01–04 · Comparing, composing, refining, returning

We supply an oriented affine plane and comparison by translation. Tip-to-tail composition is vector addition in that plane. Refinement samples a prescribed differentiable path; finite drawings do not prove a limit or smoothness. A supplied reversible rotation gives cyclic relative position. That is geometric phase, not a quantum amplitude. Colours distinguish marks; they do not specify a physical ontology.

05 · The extra structure that makes Hamilton possible

For the remaining strips we choose a state plane with dimensionless conjugate coordinates q,p and oriented area $\omega = dq \wedge dp$. This pairing is an assumption: perpendicular ink strokes do not establish it. With physical units, $p\,dq$ has units of action; momentum need not equal velocity. The patch is the object being transformed, not a background container. Its orientation matters.

05 → 06 · The local derivation

Assume an autonomous C^1 field $X=(u,v)$, its differentiable local flow, and preservation of ω for EVERY sufficiently small cell. One sheared patch only illustrates this hypothesis. Area preservation gives $u_q + v_p = 0$. Hence $\alpha = \iota_X\omega = u\,dp - v\,dq$ is closed: $d\alpha = 0$. On a simply connected local patch $\alpha = dH$, so

$$\dot{q} = u = \partial H/\partial p, \qquad \dot{p} = v = -\partial H/\partial q.$$

The Hamiltonian is reconstructed up to an additive constant. This is local: topology may obstruct a globally single-valued H. Calling H physical energy requires a physical model.

06 · An exact turning rule

Choose $H=(q^2+p^2)/2$. Then $\dot{q}=p$ and $\dot{p}=-q$. The orange arrows depict instantaneous changes, not consecutive finite kicks. For a time h the exact update is

$$q' = q\cos h + p\sin h, \qquad p' = p\cos h - q\sin h.$$

It preserves both oriented area and H. The naive Euler update $(q+h\,p,\ p-h\,q)$ instead multiplies area and q^2+p^2 by $1+h^2$. Cyclic motion belongs to this example, not to every Hamiltonian flow.

07 · A different rule and its continuous symmetry

Now choose $H=p^2/2$. The exact evolution is $\Phi_t(q,p)=(q+t\,p,p)$. Shifting the starting configuration by $T_a(q,p)=(q+a,p)$ commutes with that evolution: $\Phi_t\,T_a=T_a\,\Phi_t$. This canonical translation leaves H unchanged. Strip 7 switches rules; the oscillator of strip 6 does NOT have this translation symmetry.

08 · The Noether charge and why it is conserved

The translation generator $Y=\partial q$ obeys $\iota_Y\omega=dp$, so its charge is $J=p$. Hamilton's equations give $dJ/dt=\dot{p}=-H_q=0$. In variational language, $S=\int(p\,\dot{q}-H)dt$ is invariant under constant q translations; stationarity supplies the equations and conserved momentum. The symmetry, action structure and equations jointly justify conservation. A repeated equal mark or unchanged gap alone does not prove Noether's theorem.

Limits and the remaining debt

Supplied: affine structure, oriented pairing, regularity, model choice and variational hypotheses. Derived under those assumptions: local Hamiltonian form; two exact example flows; the translation charge and its conservation. Not established: the earlier Genesis/L1 derivation, any identity between holonomy and Noether, Born probabilities, Everett branching, Bell correlations, or an ontology of the universe. Those require further constructions and evidence.